

How to know I can(not) trust my GW inference?

...

(Usually it is better not to)

Solutions guide

How to trust GW inference a bit more



The trusting formula(s)

Double-check ini files



- Are the minimum and maximum frequencies correct?
- Is the duration correct?
- Are the detector names and data correct?
- Priors/injection values?
- Waveform arguments?
- Run settings?
- Trigger time?

⇒ Typos and mistakes can be everyone, always good to check this first before trying to debug a run.

The trusting formula(s)

Double-check ini files

*Change samplers
and their settings*

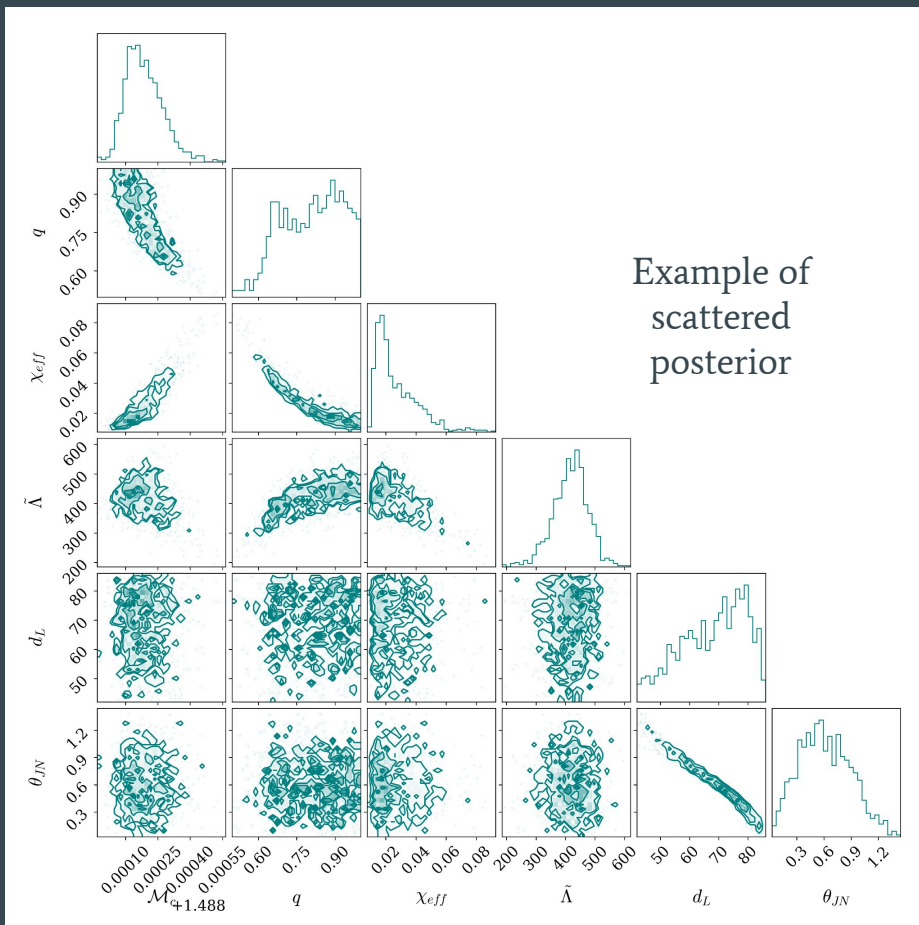


- If you see “weird” features in the posteriors (scattered, bimodality, very peaked despite low SNR, not converged...)
=> sampler stuck, struggles to cover the whole parameter space or to find the likelihood peaks

The trusting formula(s)

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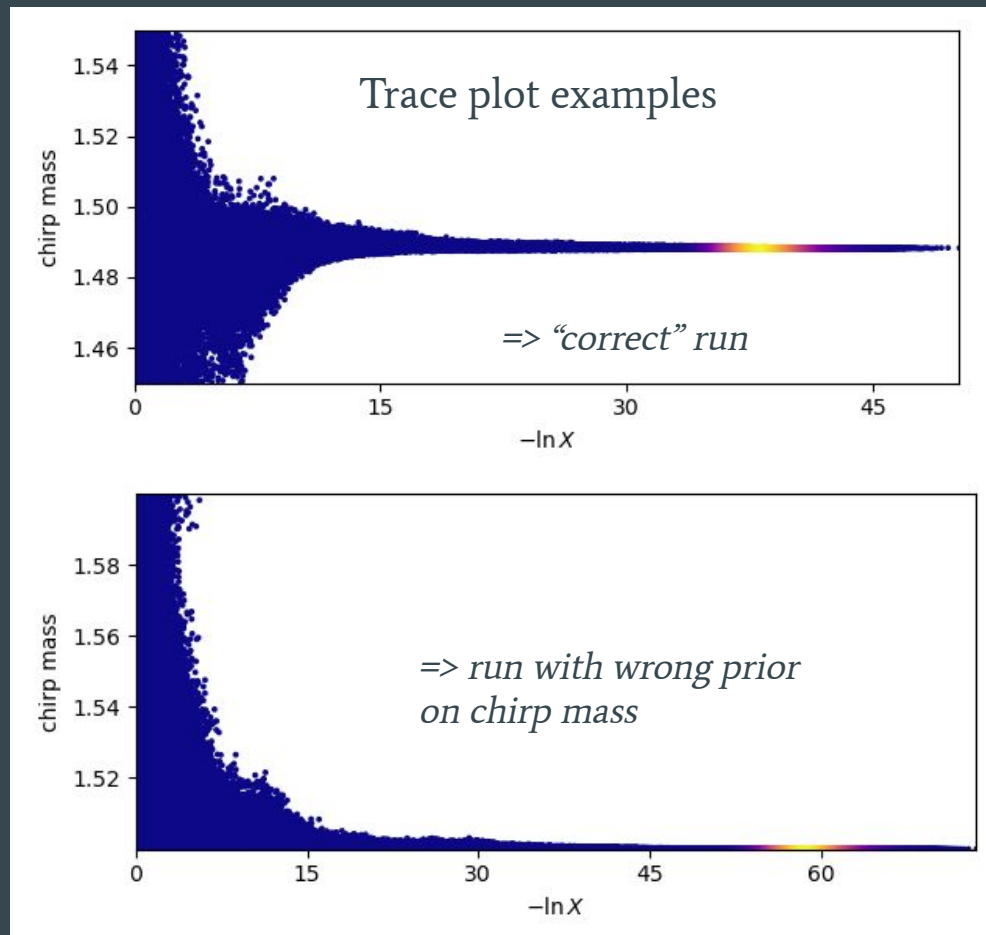


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- Trace plots to see where the sampler is going

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- Trace plots to see where the sampler is going
- **nested samplers:** usually increasing n_{live} gives a better convergence/space coverage, especially for loud signals, but there are many different settings to potentially tweak
- one could also try different samplers (for example, I would not trust pymultinest in many-parameters problems)
- **mcmc samplers:** increase the number of chains or temperatures, check single chains traceplots to see if they converge to the same region, autocorrelation time to make sure the sampler is collecting independent samples

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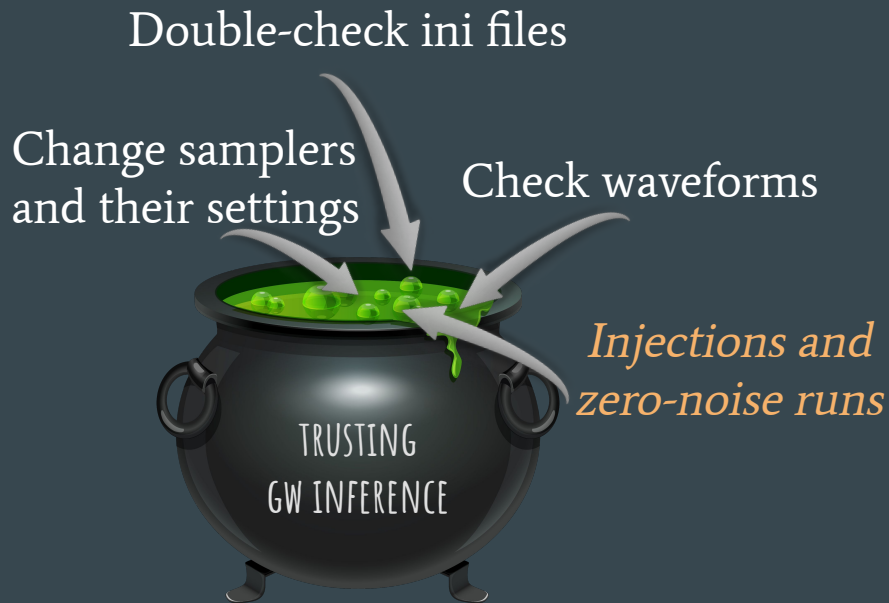
Change samplers
and their settings

Check waveforms



- Are we using the correct waveform model? (source type, aligned/precessing...)
- If we recover a large mass ratio it might be worth trying a model with higher-order mode content
- Check the waveform settings (e.g., waveform generation frequency, mode content...)
- Check / keep in mind parameters degeneracies (if your spin is off, also your mass will likely be)
- Keep in mind potential waveform limitations (e.g., BNSs with large spin, so if you recover large spins the results might not completely trustworthy)
- One could also try a different approximant to make sure the issue is not the model

The trusting formula(s)



Great debugging tools!

- If you are analyzing injections and do not recover the injected values, it is always always trying a zero-noise injection to make sure it is not a noise artifact (and in case use a different noise seed later)
- Usually noise issues do not give huge biases, but it is worth checking especially if the chirp mass is not correctly recovered
- When analyzing real data, one can perform injections with the maximum likelihood recovered values, or with the posteriors medians, for each parameter to see how a similar signal would behave and debug the analysis settings in a controlled way

The trusting formula(s)

Double-check ini files

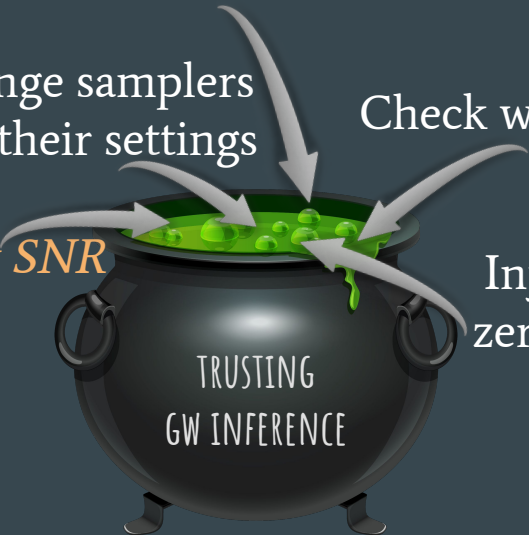
Change samplers
and their settings

Check waveforms

Injections and
zero-noise runs

Check SNR

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Double-check your signal SNR because

- if it is too low, it is very unlikely that we can constrain things well (at least 12 to have some tidal constraints)
- if it is very high, we might start seeing systematics

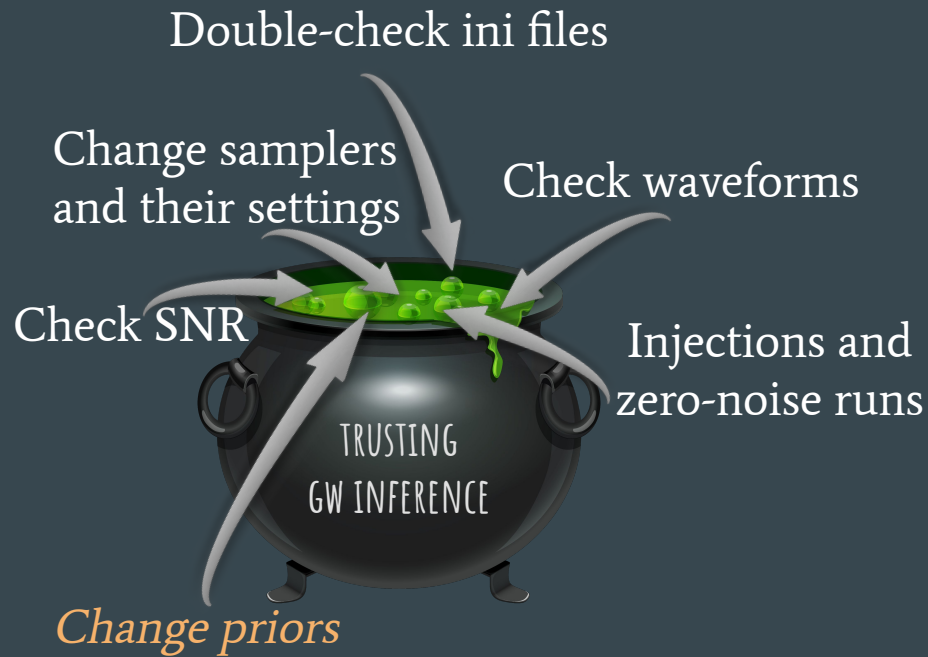


Low-SNR can be tricky for parameters that are combinations of other parameters, e.g. λ , χ_{eff} , χ_p ...

We can compare posteriors and priors for the various parameters to see if we are actually getting information

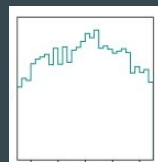
- visual comparison
- Jensen-Shannon divergence

The trusting formula(s)



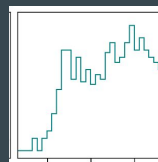
If you see railing against prior ranges

- only push the prior if it still makes physical sense (e.g., never $q > 1$ anyway by definition)
- if the SNR is too low, the data are probably just not informative enough



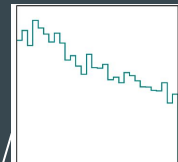
chirp mass

⇒ railing
is a
problem



mass ratio

⇒ railing at 1 is
not a
problem
(physical)

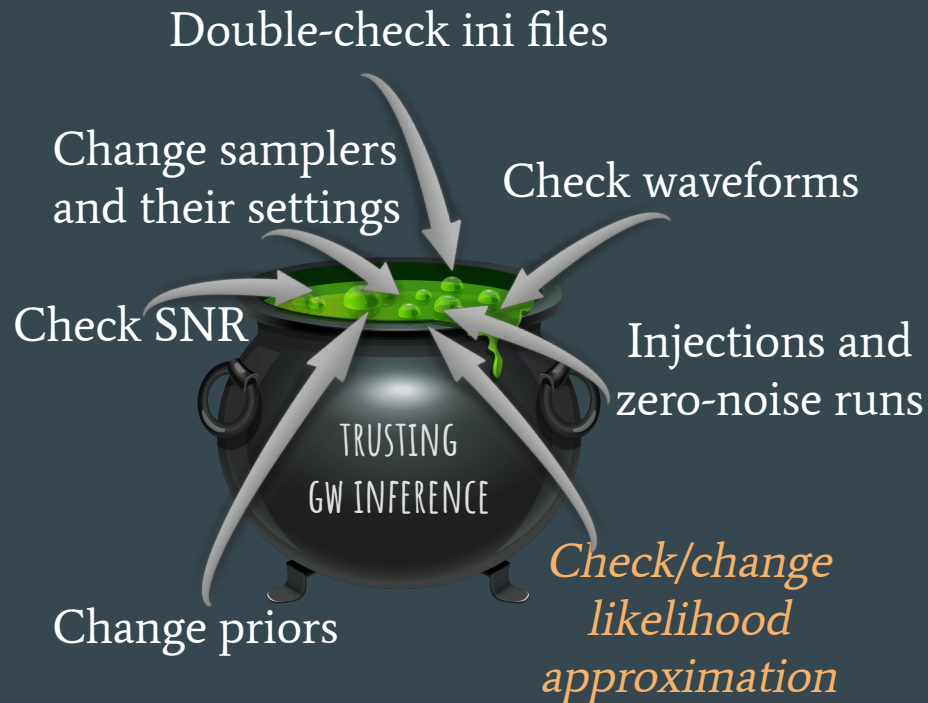


lambda

⇒ railing at 0
physical

upper bound railing:
increase prior but
check SNR

The trusting formula(s)



Likelihood approximations (multibanding, relative binning, ROQs) usually work pretty well. If you have doubts:

- check their settings (e.g., reference chirp mass for multibanding, fiducial parameters for relative binning, chirp mass bin for ROQs)
- increase accuracy (accuracy factor in multibanding, epsilon in relative binning)
- careful with higher-order models (not sure relative binning works, in multibanding one can decide how many modes to take into account)
- use coherent settings (e.g., one cannot use time marginalization with multibanding)
- a good and easy check is to compare the approximated likelihood with the 'full' GWTransient likelihood for a subset of the posterior samples

The trusting formula(s)

Double-check ini files

Change samplers
and their settings

Check waveforms

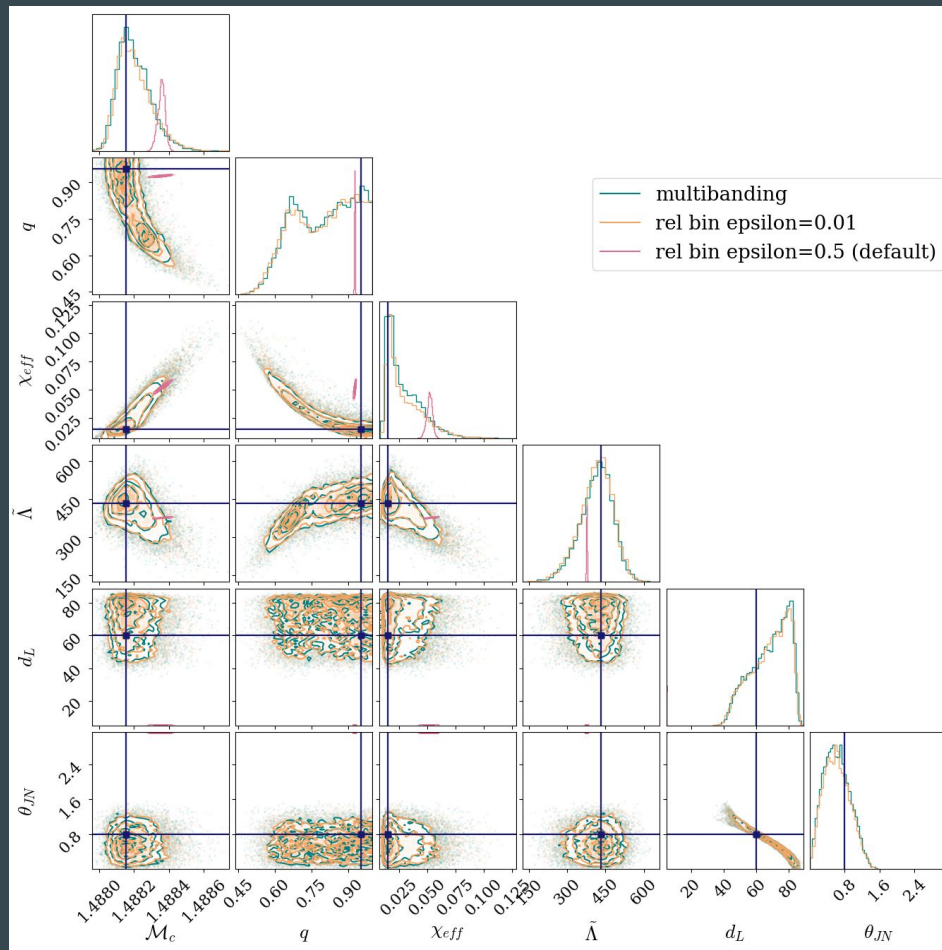
Check SNR

Injections and
zero-noise runs

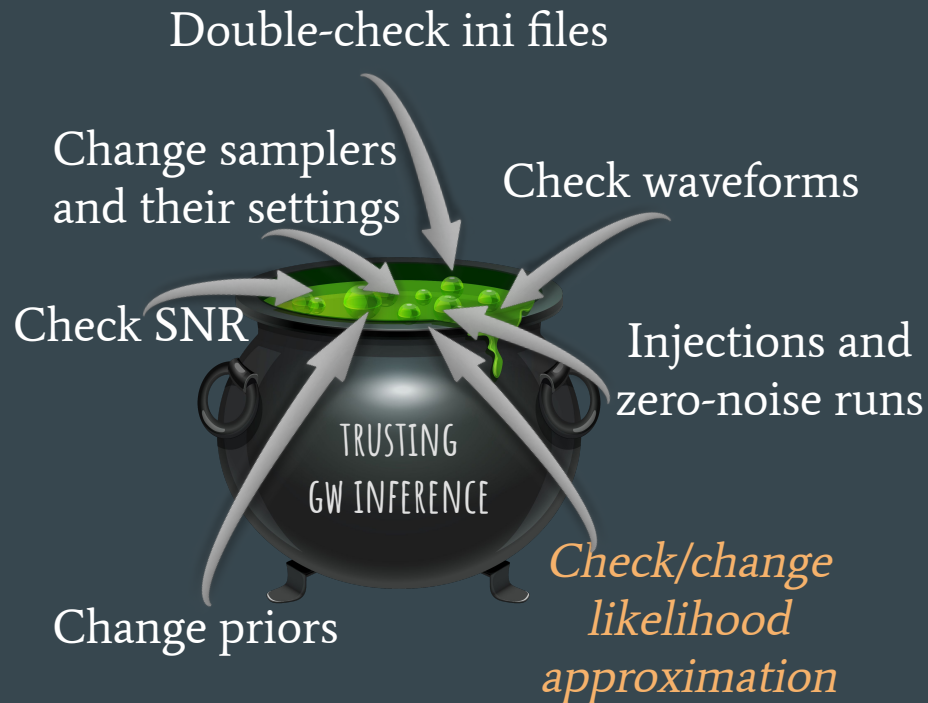
Change priors

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Check/change
likelihood
approximation



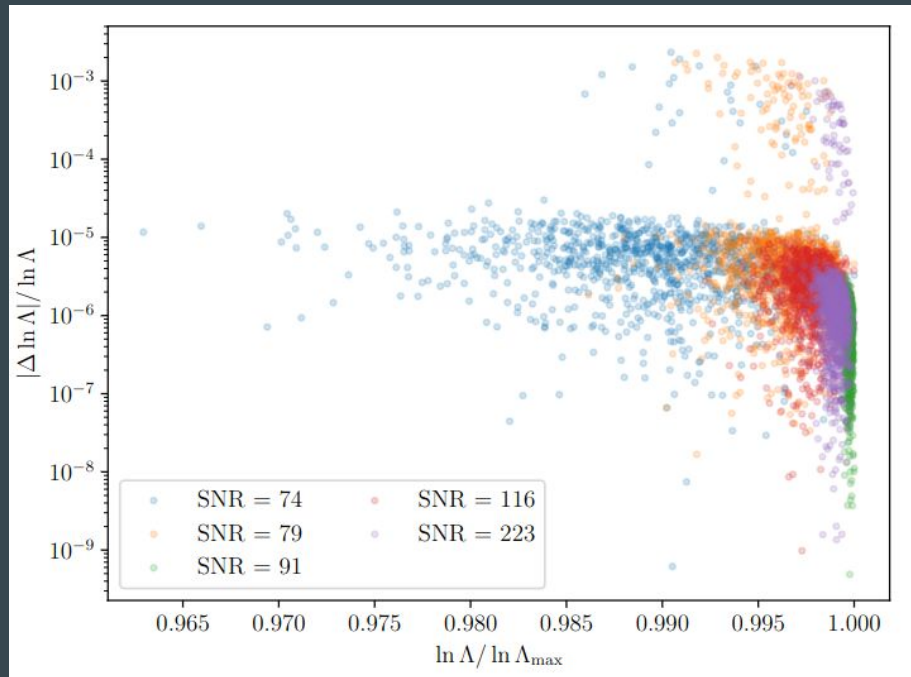
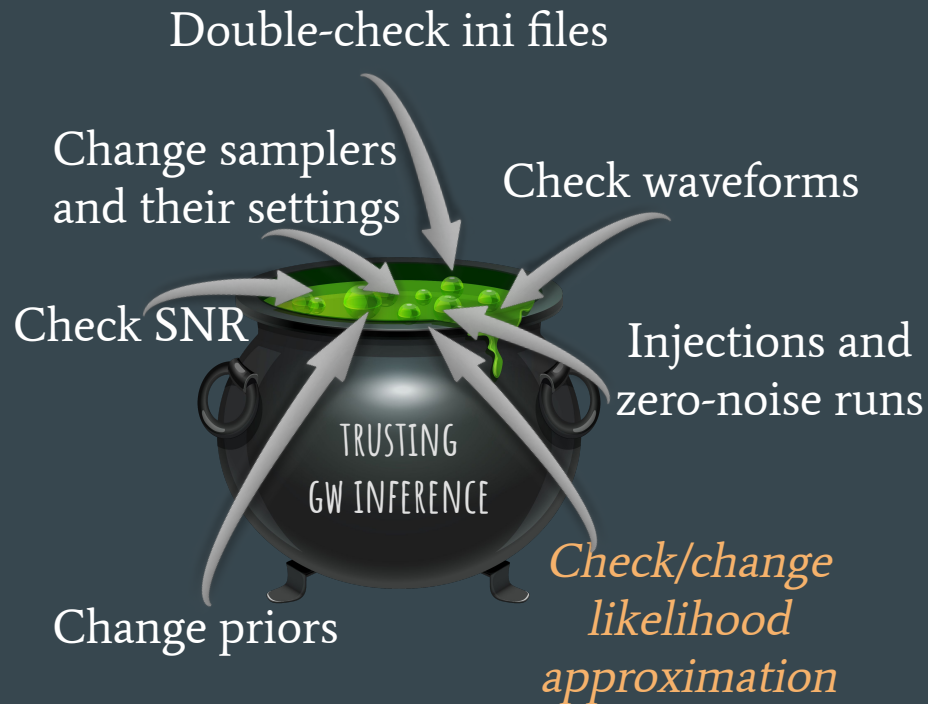
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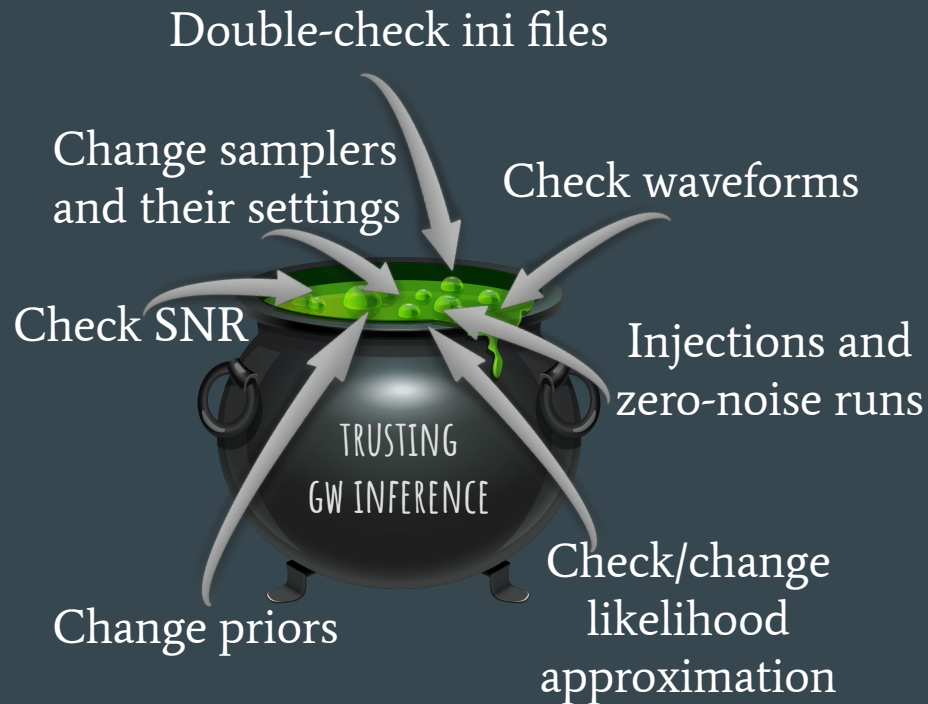
The trusting formula(s)



Example of multibanding likelihood comparison

[credits to Adrian, *Phys.Rev.Lett.* 134 (2025) 21, 211401]

The trusting formula(s)



GW inference:
many potential issues

....but also many tools to
understand/trust results

Enjoy your GW inference and
good luck ✨